

CSCI 1470 FINAL PROJECT - CHECKIN #2 REFLECTION

INTRODUCTION

The aim of the project is to learn optimal minimal-thrust target orbits for the 3-body problem via multi-agent RL. Each of the bodies is represented through an independent RL agent, whose action space is the continuous space of bounded-norm acceleration on any direction, that is, for agent i with policy π_{θ_i} , an action sampled from this policy is of the form,

$$a_i \in \mathbb{R}^3 \text{ with } \|a_i\| < B$$

where B is some global bound, chosen as a hyperparameter. For stability sake, whatever action is chosen by the agent, it must be applied for ΔT time. Therefore, the goal of each of the agents is to learn the optimal a_i to be applied for ΔT time every ΔT time-step in order to stay in some target orbit, defined as a collection of position and velocity samples for that body in the target object. This target orbit is to be reached while minimizing fuel consumption, which can be thought of as a function of the accelerations.

In order to maximize the performance of the agents, we take a Centralized Training Decentralized Training approach to our system. Where the critic and reward function will be global, while each of the agents gets specific individual agent information. To achieve the goals outlined earlier, the reward function will be of the form,

$$R = \omega_{\text{orbit}_x} R_{\text{orbit}_x} + \omega_{\text{orbit}_v} R_{\text{orbit}_v} + \omega_{\text{fuel}} R_{\text{fuel}} + \omega_{\text{survive}} R_{\text{survive}}$$

with $R_{\text{orbit}_x} = \sum_{i=1}^3 \|x_i - x_i^*\|^2$, $R_{\text{orbit}_v} = \sum_{i=1}^3 \|v_i - v_i^*\|^2$, $R_{\text{fuel}} = \ln\left(\frac{\beta}{\sum_{i=1}^3 \|a_i\|^2}\right)$ and R_{survive} adds 0.1 for every time-step that there is no collision and -1000 if there is a collision. Here x_i^* and v_i^* are the positions and velocities for a given point in the target orbit (the ones that are closest to the current bodies positions), and β is a bound for the optimal maximum amount of fuel to be consumed by the system for a specific time-step.

CHALLENGES

The first and biggest of challenges was how to encode the target orbit. In our system, the three bodies are set to some specific fixed points at the beginning, these can be tuned by the user, and the agent's must learn what thrust's to apply so that all three bodies move synchronously in our target orbit, say the figure-8 orbit. The first naive approach could be to take this samples by time-steps and then compare the current positions of our bodies to the corresponding ones in the orbit for that specific orbit, but then this requires that we enter the orbit exactly at some multiple of the period of the orbit, meaning that if it misses the exact time, it has to be flying around wasting fuel until the next multiple of the period. Another approach, could be to sample a bunch of points in the orbit with their corresponding velocities (which would help capture orientation and continuity) and then for each of the body measure the distance to it's closest point in the orbit, however, this is troublesome as for all three bodies to be in the orbit, they must be synchronized, together, if the points in the orbit that are closest to each point independently come from different time samples, we are not achieving this synchronization and we might not be in the correct orbit. The solution we came up with was finding the sample that was globally closes to the three bodies, and compute the error based on that one, that is for a given time t in our simulation, we find,

$$k(t) = \operatorname{argmin}_{t'} \left(\sum_{i=1}^3 \|x_i(t) - x_i^*(t')\|^2 \right)$$

and then compute,

$$R_{\text{orbit}_x}(t) = \sum_{i=1}^3 \|x_i(t) - x_i^*(k(t))\|^2, \quad \text{and} \quad R_{\text{orbit}_v}(t) = \sum_{i=1}^3 \|v_i(t) - v_i^*(k(t))\|^2$$

INSIGHTS

We are currently starting the training and experimentation process and expect that the weight for the fuel reward will be fairly small, as given a target action time-step ΔT we believe the physics of the system will naturally favor smaller actions, rather than large ones, which produce less stable orbital paths and thus render the bodies further from the target orbit.

One aspect we are interested in finding out more about through testing is the step-size of the simulation, which is different from the action step-size ΔT . In previous literature, learned step-sized through RL approaches have leveraged stable accurate simulation in more efficient times, as opposed to earlier approaches where the simulation step-size is fixed to a really small δt , which is our current approach. We wonder, whether a variable step-size approach would greatly increase our training speed without hurting simulation accuracy enough to sacrifice training convergence.

PLAN

Our current plan is to batch train on a different set of hyperparameters, some that favor closeness to the target orbit as opposed to fuel consumption and vice-versa. Moreover, in our current approach, the information each of the agents receives comes from a GNN where each node is one of the bodies (with their corresponding position, velocity and mass) and their edges encoding their distance position difference, which generates three embeddings, one for each agent. Our current edge's only encode $(x_i - x_j, \|x_i - x_j\|)$, however, we are wondering whether adding a third parameter the difference between center of mass distance and the sum of their corresponding radii would help encode distance to collision better, and in turn reduce collision between the bodies.

We also expect to map out a set of stability-prone initial positions and optimal-max fuel parameters for different mass and initial position configurations.